CLOUD INFRASTRUCTURE AUTOMATION USING TERRAFORM AND AWS

Services: EC2, RDS, VPC, Terraform

Description: Automate the provisioning of an application stack with Terraform. The stack includes EC2 instances in a VPC with a public and private subnet, a load balancer, and an RDS database.

Skills: Infrastructure as Code (IaC), Terraform, VPC setup.

Definition: This project automates the deployment of AWS cloud infrastructure using Terraform. It creates resources including a VPC, subnets, EC2 instances, an Application Load Balancer, and an RDS database, enabling infrastructure management through code rather than manual setup.

Scope of the Project: The scope of this project is to design and deploy a secure and scalable AWS cloud infrastructure using Terraform. It includes setting up networking components, compute resources, load balancing, and a database environment. The project demonstrates how Infrastructure as Code (IaC) can automate resource provisioning, reduce manual effort, and ensure consistent infrastructure deployment.

Methodologies:

1. **Infrastructure as Code (IaC) using Terraform – Best Method**

- Uses Terraform to automate AWS infrastructure deployment.
- Infrastructure is defined and managed through code.
- Supports version control and reusable configurations.
- Reduces deployment time and configuration errors.
- Enables repeatable and consistent infrastructure creation.
- Suitable for enterprise and production environments.

2. **Manual Configuration using AWS Console – Alternative Method**

- Resources are created manually through the AWS Management Console.
- Suitable only for small-scale environments.
- Requires significant manual effort.
- Difficult to maintain consistency across environments.
- Prone to configuration mistakes and operational overhead.
- Not suitable for large-scale infrastructure deployments.

Services Used in this Project:

1. Amazon EC2 (Elastic Compute Cloud)
Definition: Amazon EC2 is a cloud service that provides virtual servers to run applications in the AWS environment.
Purpose: EC2 is used to host and run the web application inside the VPC.
Role: EC2 acts as the application server that receives traffic from the load balancer and processes user requests.

2. RDS (Relational Database Service)

Definition: Amazon RDS is a managed database service used to create, operate, and scale relational databases in the AWS cloud.

Purpose: RDS is used to store and manage the application's data securely in the private subnet.

Role: RDS acts as the backend database that stores and retrieves data for the application running on the EC2 instance.

3. VPC (Virtual Private Cloud)

Definition: Amazon VPC is a virtual private network in AWS that allows you to securely host and manage cloud resources.

Purpose: VPC is used to create an isolated and secure network environment for deploying the application infrastructure.

Role: VPC acts as the main network that connects and controls communication between EC2, subnets, load balancer, and RDS.

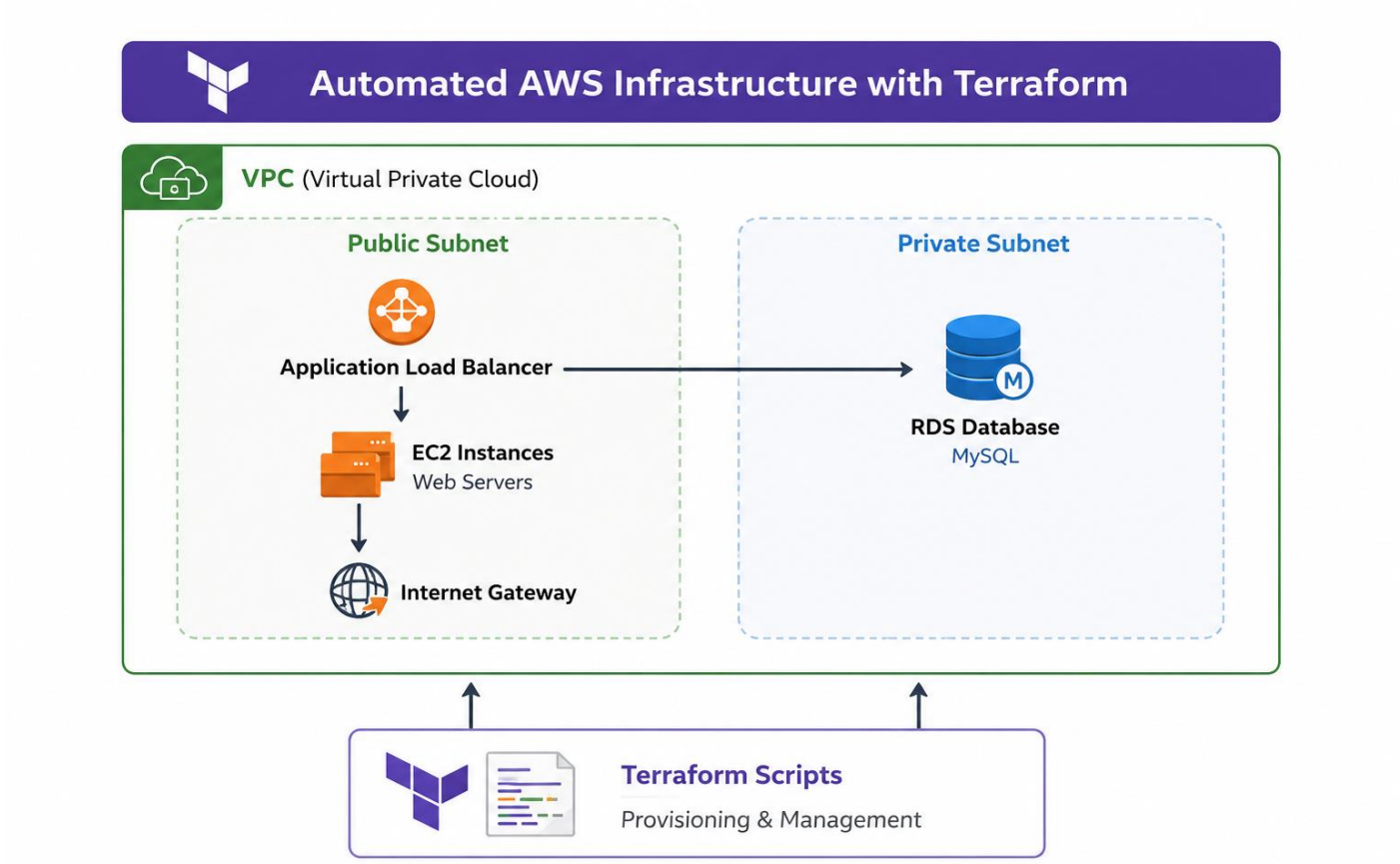
4. Terraform

Definition: Terraform is an Infrastructure as Code (IaC) tool used to create and manage cloud resources using configuration files.

Purpose: Terraform is used to automate the provisioning of AWS infrastructure such as VPC, EC2, load balancer, and RDS.

Role: Terraform acts as the automation tool that deploys and manages the entire application infrastructure through code.

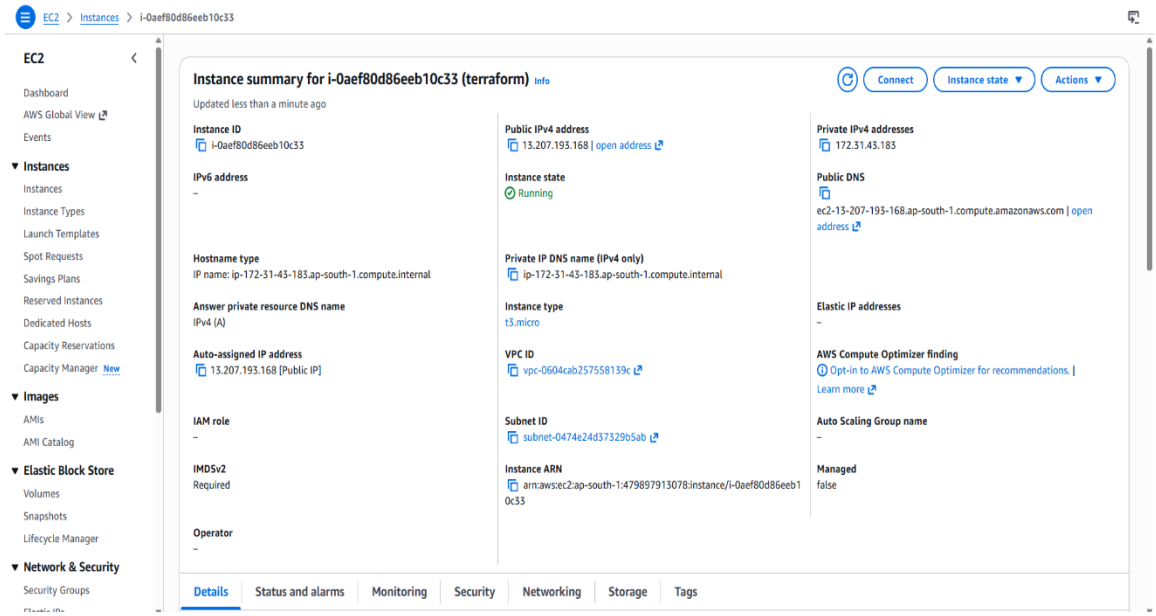
Architecture Diagram:



Implementation

STEP 1: Launch an EC2 Server

- Go to AWS Console → EC2 → Launch Instance.
- Choose Amazon Linux 2023 AMI.
- Instance Type: t3. micro.
- Storage: 20 GB.
- Region: Mumbai (ap-south-1).
- Configure Security Group.
- Create or Select Key Pair.
- Launch Instance.



STEP :2 Install and Check the Terraform version

- Connect to the EC2 instance.
- Download Terraform package.
- Install Terraform.
- Verify installation:
- terraform -version

```
[ec2-user@ip-172-31-43-183 ~]$ terraform -version
Terraform v1.15.5
on linux_amd64
[ec2-user@ip-172-31-43-183 ~]$ aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.1.172-216.329.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-172-31-43-183 ~]$ aws configure
AWS Access Key ID [****************2R5I]: AWS Secret Access Key [****************dPUC]: Default region name [ap-south-1]: Default output format [None]: [ec2-user@ip-172-31-43-183 ~]$
[ec2-user@ip-172-31-43-183 ~]$ mkdir terraform-aws-project
[ec2-user@ip-172-31-43-183 ~]$ cd terraform-aws-project/
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ cd ..
[ec2-user@ip-172-31-43-183 ~]$ ls
terraform-aws-project
[ec2-user@ip-172-31-43-183 ~]$ cd terraform-aws-project/
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ ls
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ nano main.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ ls
main.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ nano variable.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ ls
main.tf  variable.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ nano terraform.tfvars
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ ls
main.tf  terraform.tfvars  variable.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ nano output.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ ls
main.tf  output.tf  terraform.tfvars  variable.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.47.0...
- Installed hashicorp/aws v6.47.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

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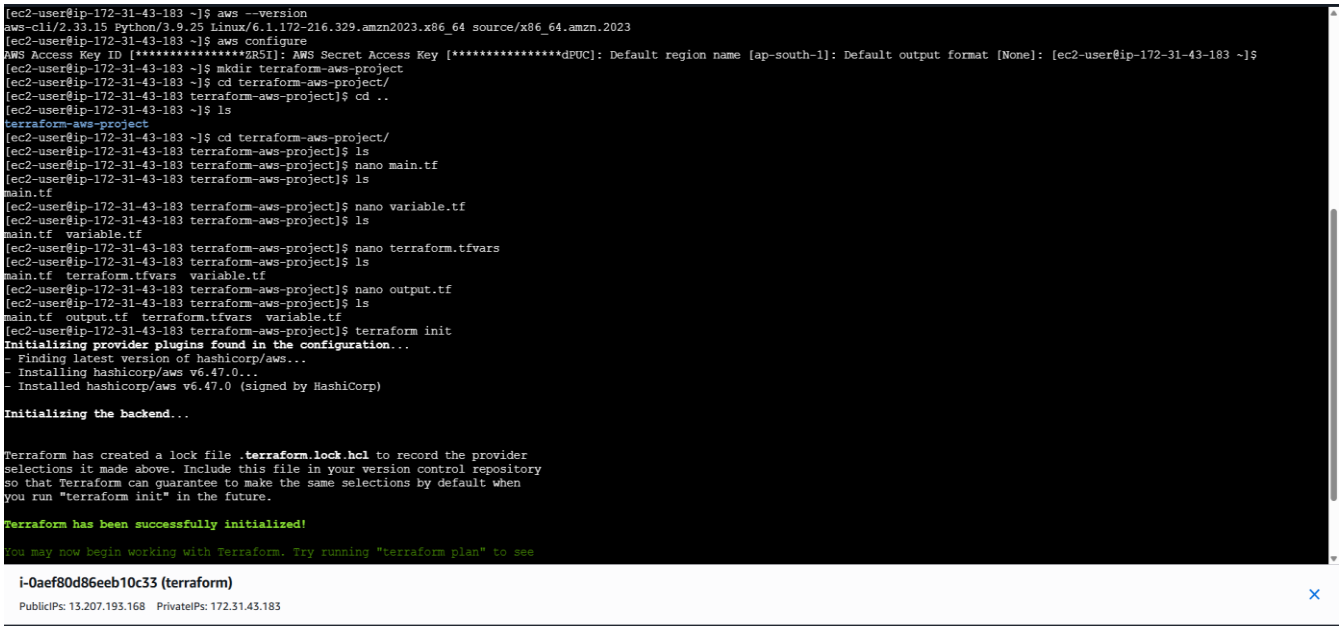
STEP3: Check aws version

- Check AWS CLI version:
- aws --version



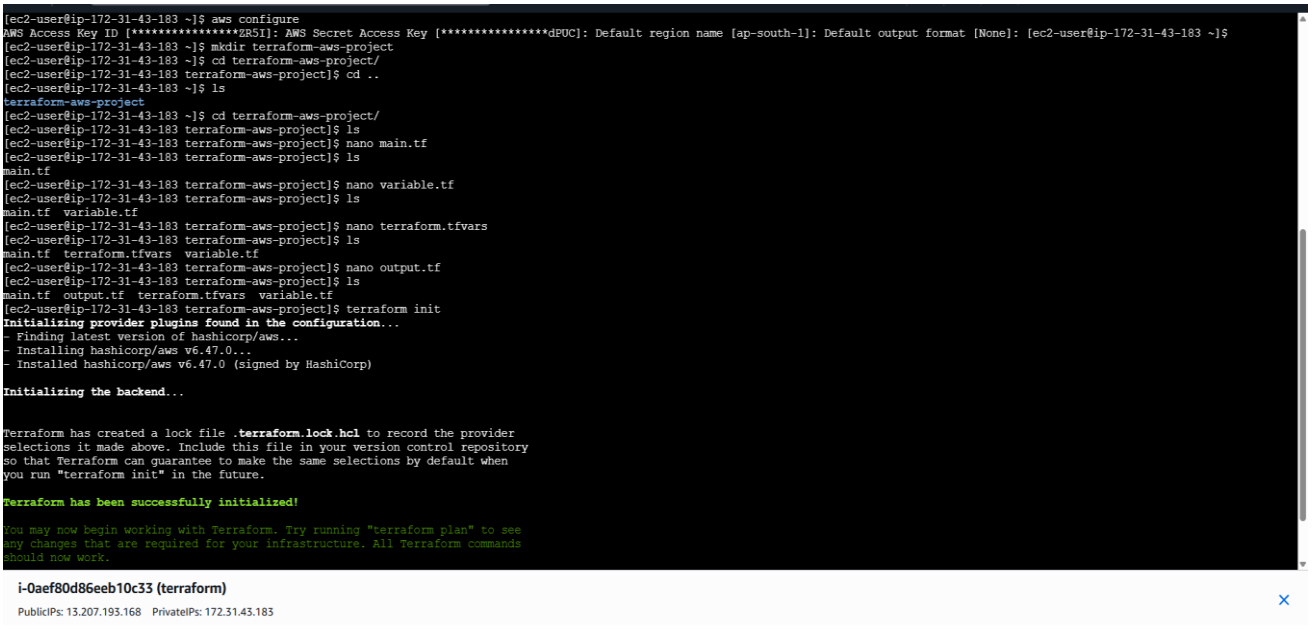
STEP 4: Create Access Keys and Configure AWS CLI

- Go to IAM → Users.
- Create Access Key.
- Configure AWS CLI:

aws configure

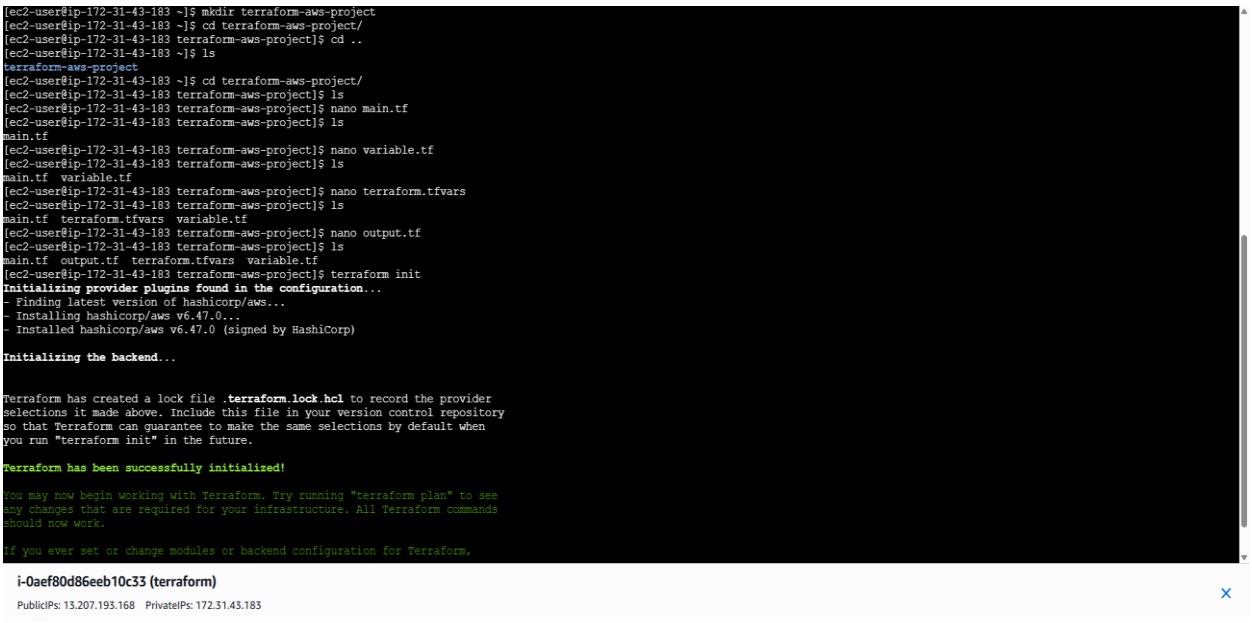
- **Enter:**

- Access Key ID
- Secret Access Key
- Region
- Output Format



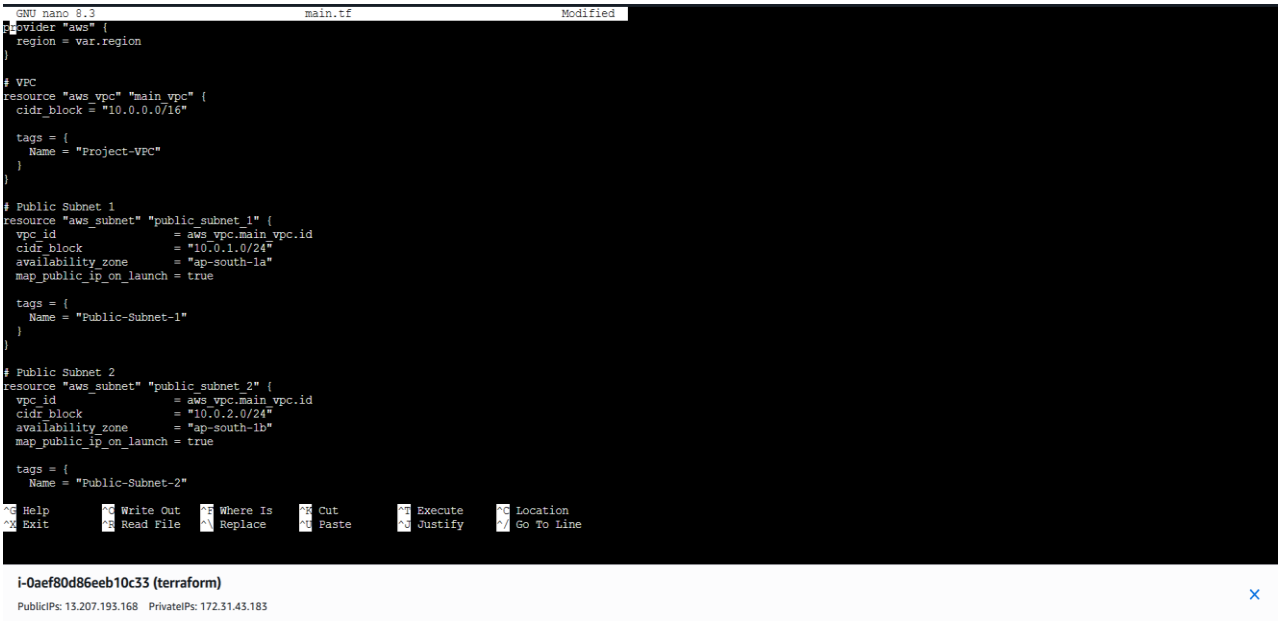
STEP 5: Created a project directory and added four Terraform file .Create the project folder and Terraform files:

- main.tf
- variables.tf
- terraform.tfvars
- outputs.tf



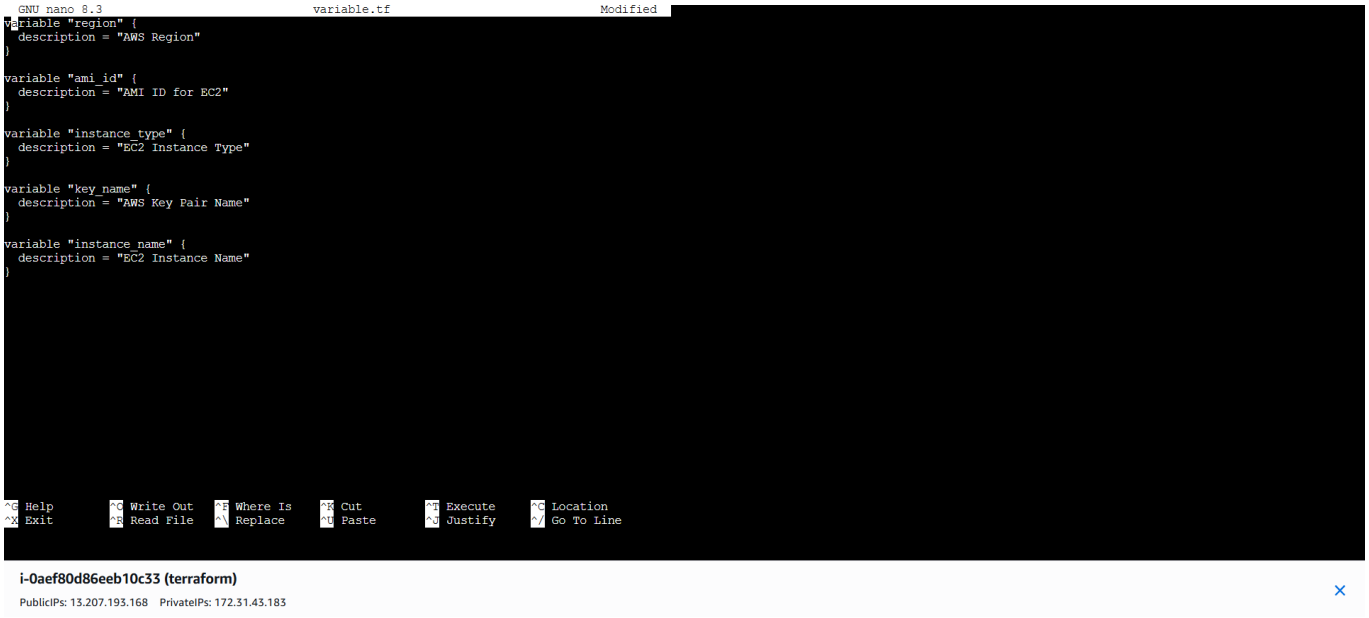
STEP 6: main.tf

- Configure VPC.
- Configure Subnets.
- Configure Internet Gateway.
- Configure Route Tables.
- Configure Security Groups.
- Configure EC2 Instance.
- Configure Application Load Balancer.
- Configure RDS Database.



STEP 7: variable.tf

- Define Terraform variables.
- Store reusable configuration values.

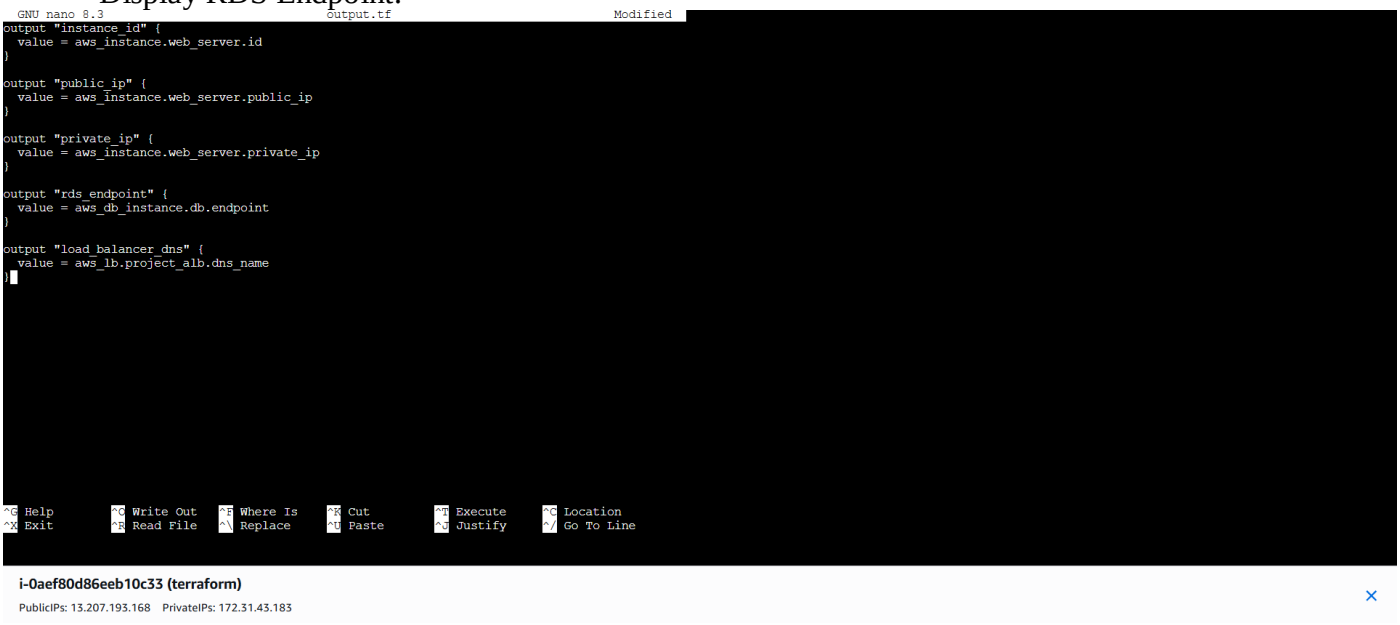


STEP 8: terraform. tfvars

- Assign values to variables.
- Customize infrastructure settings.

STEP 9: output.tf

- Display resource IDs.
- Display Load Balancer DNS.
- Display RDS Endpoint.



STEP 10: Initialize Terraform

- Run:
- terraform init
- Downloads required providers.
- Initializes Terraform workspace.

```
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.47.0...
- Installed hashicorp/aws v6.47.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform fmt
main.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform validate
Success! The configuration is valid.

[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_db_instance.db will be created
  + resource "aws_db_instance" "db" {
      + address                               = (known after apply)
      + allocated_storage                     = 20
```

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STEP 11: Format Terraform Configuration

- Run
- terraform fmt
- Formats Terraform files.

```
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform fmt
main.tf
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform validate
Success! The configuration is valid.

[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_db_instance.db will be created
  + resource "aws_db_instance" "db" {
      + address                               = (known after apply)
      + allocated_storage                     = 20
      + apply_immediately                     = false
      + arn                                   = (known after apply)
      + auto_minor_version_upgrade            = true
      + availability_zone                     = (known after apply)
      + backup_retention_period               = (known after apply)
      + backup_target                         = (known after apply)
      + backup_window                         = (known after apply)
      + ca_cert_identifier                    = (known after apply)
      + character_set_name                    = (known after apply)
      + copy_tags_to_snapshot                 = false
      + database_insights_mode                = (known after apply)
      + db_name                               = "projectdb"
      + db_subnet_group_name                  = "project-db-subnet-group"
      + dedicated_log_volume                  = false
      + delete_automated_backups              = true
      + domain_fqdn                           = (known after apply)
      + endpoint                              = (known after apply)
      + engine                                = "mysql"
      + engine_lifecycle_support              = (known after apply)
      + engine_version                        = "8.0"
      + engine_version_actual                 = (known after apply)
      + hosted_zone_id                        = (known after apply)
      + id                                    = (known after apply)
```

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STEP 12 : Validate Terraform Configuration

- Run:
- terraform validate
- Checks configuration syntax and dependencies.

```
ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform validate
uccess! The configuration is valid.

ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform plan

erraform used the selected providers to generate the following execution plan. Resource
ctions are indicated with the following symbols:
  + create

erraform will perform the following actions:

  # aws_db_instance.db will be created
  + resource "aws_db_instance" "db" {
      + address                               = (known after apply)
      + allocated_storage                     = 20
      + apply_immediately                     = false
      + arn                                   = (known after apply)
      + auto_minor_version_upgrade            = true
      + availability_zone                     = (known after apply)
      + backup_retention_period               = (known after apply)
      + backup_target                         = (known after apply)
      + backup_window                         = (known after apply)
      + ca_cert_identifier                    = (known after apply)
      + character_set_name                    = (known after apply)
      + copy_tags_to_snapshot                 = false
      + database_insights_mode                = (known after apply)
      + db_name                               = "projectdb"
      + db_subnet_group_name                  = "project-db-subnet-group"
      + dedicated_log_volume                  = false
      + delete_automated_backups              = true
      + domain_fqdn                           = (known after apply)
      + endpoint                              = (known after apply)
      + engine                                = "mysql"
      + engine_lifecycle_support              = (known after apply)
      + engine_version                        = "8.0"
      + engine_version_actual                 = (known after apply)
      + hosted_zone_id                        = (known after apply)
      + id                                    = (known after apply)
      + identifier                            = "projectdb"
      + identifier_prefix                     = (known after apply)
```

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STEP 13: Check Infrastructure Plan

- Run:
- terraform plan
- Displays resources that will be created.

```
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_db_instance.db will be created
  + resource "aws_db_instance" "db" {
      + address                               = (known after apply)
      + allocated_storage                     = 20
      + apply_immediately                     = false
      + arn                                   = (known after apply)
      + auto_minor_version_upgrade            = true
      + availability_zone                     = (known after apply)
      + backup_retention_period               = (known after apply)
      + backup_target                         = (known after apply)
      + backup_window                         = (known after apply)
      + ca_cert_identifier                    = (known after apply)
      + character_set_name                    = (known after apply)
      + copy_tags_to_snapshot                 = false
      + database_insights_mode                = (known after apply)
      + db_name                               = "projectdb"
      + db_subnet_group_name                  = "project-db-subnet-group"
      + dedicated_log_volume                  = false
      + delete_automated_backups              = true
      + domain_fqdn                           = (known after apply)
      + endpoint                              = (known after apply)
      + engine                                = "mysql"
      + engine_lifecycle_support              = (known after apply)
      + engine_version                        = "8.0"
      + engine_version_actual                 = (known after apply)
      + hosted_zone_id                        = (known after apply)
      + id                                    = (known after apply)
      + identifier                            = "projectdb"
      + identifier_prefix                     = (known after apply)
      + instance_class                        = "db.t3.micro"
      + iops                                  = (known after apply)
      + kms_key_id                            = (known after apply)
```

i-0aef80d86eeb10c33 (terraform)

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```
  + resource "aws_vpc" "main_vpc" {
      + arn                                  = (known after apply)
      + cidr_block                           = "10.0.0.0/16"
      + default_network_acl_id               = (known after apply)
      + default_route_table_id               = (known after apply)
      + default_security_group_id            = (known after apply)
      + dhcp_options_id                      = (known after apply)
      + enable_dns_hostnames                 = (known after apply)
      + enable_dns_support                   = true
      + enable_network_address_usage_metrics = (known after apply)
      + id                                   = (known after apply)
      + instance_tenancy                     = "default"
      + ipv6_association_id                  = (known after apply)
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + region                               = "ap-south-1"
      + tags                                 = {
          + "Name" = "Project-VPC"
        }
      + tags_all                             = {
          + "Name" = "Project-VPC"
        }
    }

Plan: 16 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_id      = (known after apply)
  + load_balancer_dns = (known after apply)
  + private_ip       = (known after apply)
  + public_ip        = (known after apply)
  + rds_endpoint     = (known after apply)


Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take
exactly these actions if you run "terraform apply" now.
[ec2-user@ip-172-31-43-183 terraform-aws-project]$
```

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STEP 14: Deploy Infrastructure

- Run: terraform apply
- Review execution plan.
- Type: yes
- Deploy resources.

```
[ec2-user@ip-172-31-43-183 terraform-aws-project]$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_db_instance.db will be created
  + resource "aws_db_instance" "db" {
      + address                               = (known after apply)
      + allocated_storage                     = 20
      + apply_immediately                     = false
      + arn                                   = (known after apply)
      + auto_minor_version_upgrade            = true
      + availability_zone                     = (known after apply)
      + backup_retention_period               = (known after apply)
      + backup_target                         = (known after apply)
      + backup_window                         = (known after apply)
      + ca_cert_identifier                    = (known after apply)
      + character_set_name                    = (known after apply)
      + copy_tags_to_snapshot                 = false
      + database_insights_mode                = (known after apply)
      + db_name                               = "projectdb"
      + db_subnet_group_name                  = "project-db-subnet-group"
      + dedicated_log_volume                  = false
      + delete_automated_backups              = true
      + domain_fqdn                           = (known after apply)
      + endpoint                              = (known after apply)
      + engine                                = "mysql"
      + engine_lifecycle_support              = (known after apply)
      + engine_version                        = "9.0"
      + engine_version_actual                 = (known after apply)
      + hosted_zone_id                        = (known after apply)
      + id                                    = (known after apply)
      + identifier                            = "projectdb"
      + identifier_prefix                     = (known after apply)
      + instance_class                        = "db.t3.micro"
      + iops                                  = (known after apply)
      + kms_key_id                            = (known after apply)
```

i-0aef80d86eeb10c33 (terraform)

PublicIPs: 13.207.193.168 PrivateIPs: 172.31.43.183

```
  + resource "aws_vpc" "main_vpc" {
      + arn                                  = (known after apply)
      + cidr_block                           = "10.0.0.0/16"
      + default_network_acl_id               = (known after apply)
      + default_route_table_id               = (known after apply)
      + default_security_group_id            = (known after apply)
      + dhcp_options_id                      = (known after apply)
      + enable_dns_hostnames                 = (known after apply)
      + enable_dns_support                   = true
      + enable_network_address_usage_metrics = (known after apply)
      + id                                   = (known after apply)
      + instance_tenancy                     = "default"
      + ipv6_association_id                  = (known after apply)
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + region                               = "ap-south-1"
      + tags                                 = {
          + "Name" = "Project-VPC"
        }
      + tags_all                             = {
          + "Name" = "Project-VPC"
        }
    }

Plan: 16 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_id      = (known after apply)
  + load_balancer_dns = (known after apply)
  + private_ip       = (known after apply)
  + public_ip        = (known after apply)
  + rds_endpoint     = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
```

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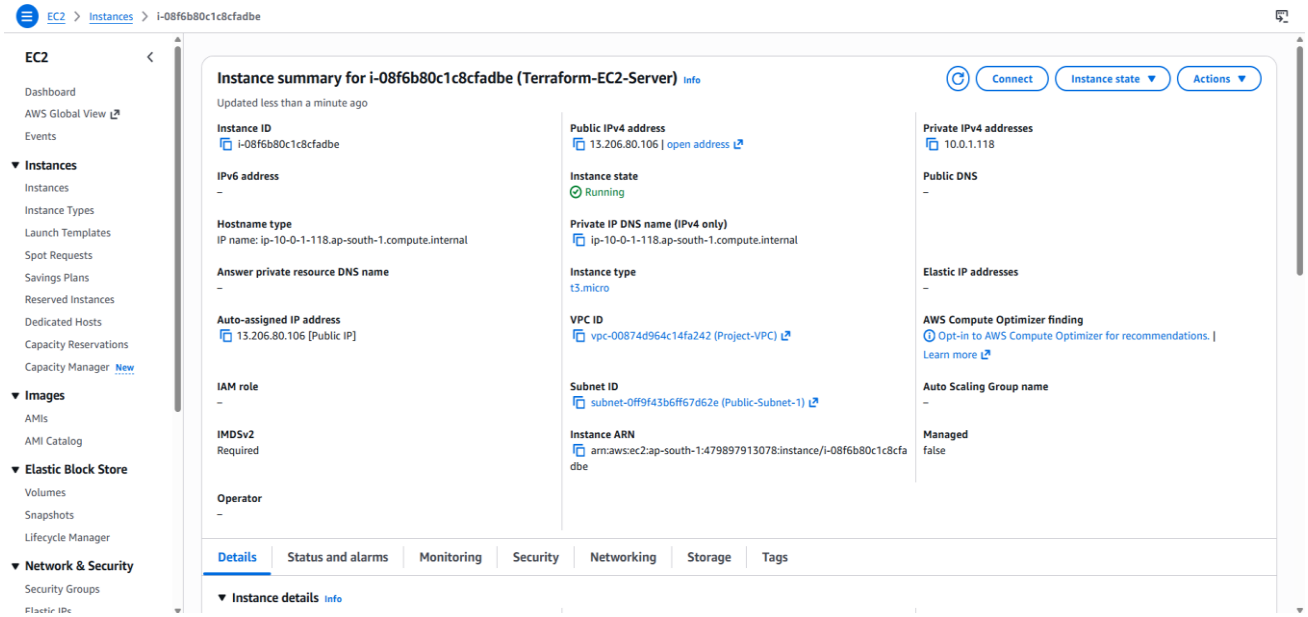
```
aws_db_instance.db: Still creating... [01m40s elapsed]
aws_lb_project_alb: Still creating... [01m50s elapsed]
aws_db_instance.db: Still creating... [01m50s elapsed]
aws_lb_project_alb: Still creating... [02m00s elapsed]
aws_db_instance.db: Still creating... [02m00s elapsed]
aws_lb_project_alb: Still creating... [02m10s elapsed]
aws_db_instance.db: Still creating... [02m10s elapsed]
aws_lb_project_alb: Still creating... [02m20s elapsed]
aws_db_instance.db: Still creating... [02m20s elapsed]
aws_lb_project_alb: Still creating... [02m30s elapsed]
aws_db_instance.db: Still creating... [02m30s elapsed]
aws_lb_project_alb: Still creating... [02m40s elapsed]
aws_db_instance.db: Still creating... [02m40s elapsed]
aws_lb_project_alb: Still creating... [02m50s elapsed]
aws_db_instance.db: Still creating... [02m50s elapsed]
aws_lb_project_alb: Creation complete after 2m51s [id=arn:aws:elasticloadbalancing:ap-south-1:479897913078:loadbalancer/app/project-alb/badaeb865dd0b860]
aws_lb_listener.listener: Creating...
aws_lb_listener.listener: Creation complete after 0s [id=arn:aws:elasticloadbalancing:ap-south-1:479897913078:listener/app/project-alb/badaeb865dd0b860/1f6b087c27b53b84]
aws_db_instance.db: Still creating... [03m00s elapsed]
aws_db_instance.db: Still creating... [03m10s elapsed]
aws_db_instance.db: Still creating... [03m20s elapsed]
aws_db_instance.db: Still creating... [03m30s elapsed]
aws_db_instance.db: Still creating... [03m40s elapsed]
aws_db_instance.db: Still creating... [03m50s elapsed]
aws_db_instance.db: Still creating... [04m00s elapsed]
aws_db_instance.db: Still creating... [04m11s elapsed]
aws_db_instance.db: Still creating... [04m21s elapsed]
aws_db_instance.db: Still creating... [04m31s elapsed]
aws_db_instance.db: Creation complete after 4m34s [id=db-JT4LVS6UVGB52T3CHVRDHYOV4E]

Apply complete! Resources: 16 added, 0 changed, 0 destroyed.

Outputs:

instance_id = "i-08f6b80c1c8cfadbe"
load_balancer_dns = "project-alb-1874621532.ap-south-1.elb.amazonaws.com"
private_ip = "10.0.1.118"
public_ip = "13.206.80.106"
rds_endpoint = "projectdb.cjaocmm6kkdm.ap-south-1.rds.amazonaws.com:3306"
[ec2-user@ip-172-31-43-183 terraform-aws-project]$
```

i-0aef80d86eeb10c33 (terraform)

PublicIPs: 13.207.193.168 PrivateIPs: 172.31.43.183

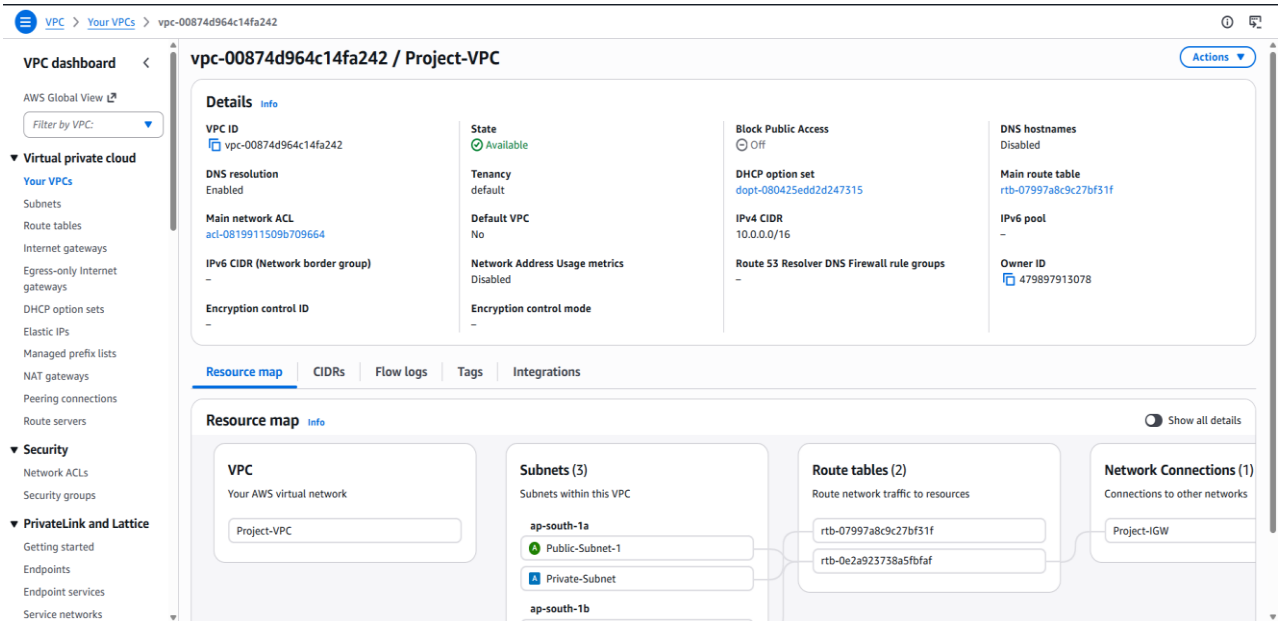
STEP 15: Verify Resources in AWS Console → EC2 (Elastic Compute Cloud)

- Open AWS Console → EC2.
- Verify EC2 instance creation.
- Check instance status.



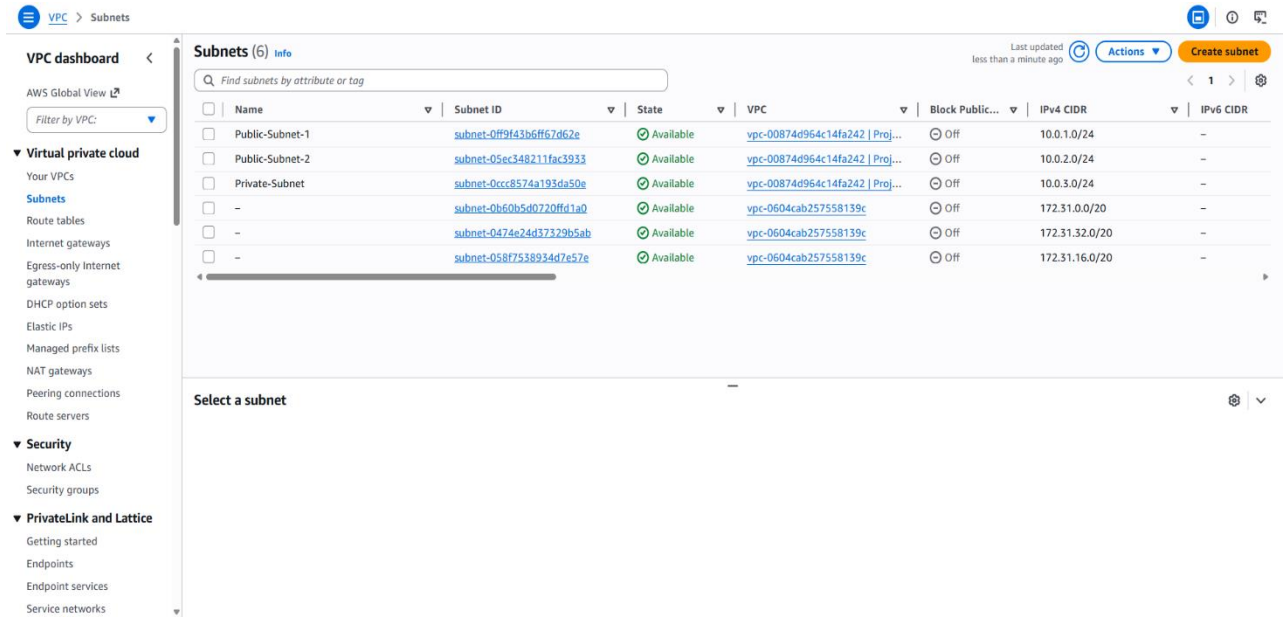
STEP 16: VPC (Virtual Private Cloud)

- Open AWS Console → VPC.
- Verify VPC creation.
- Verify networking configuration.
-



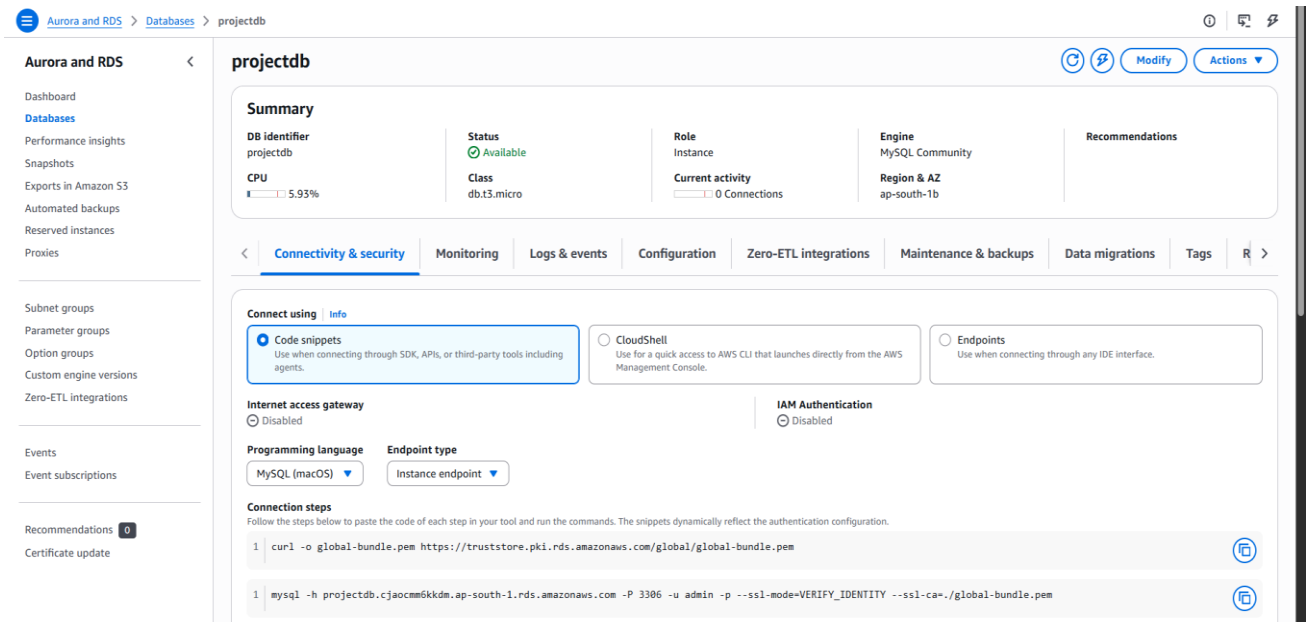
STEP 17: Public Subnet and Private Subnet

- Confirm subnet creation.
- Verify route table associations.



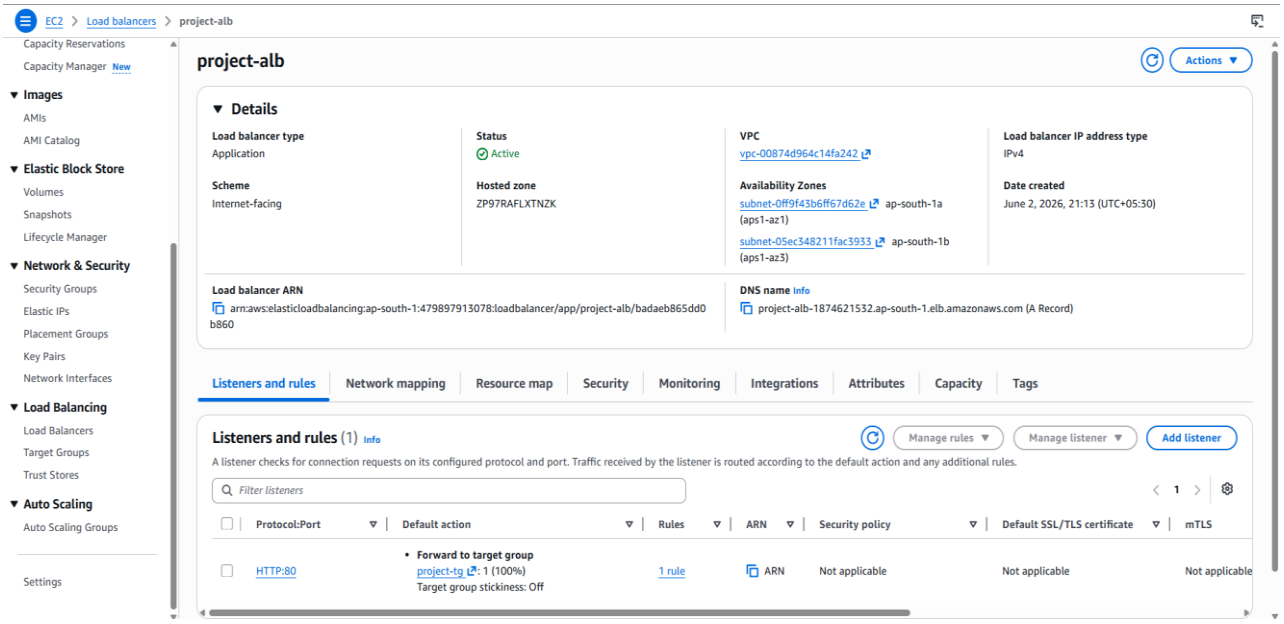
STEP 18: RDS (Relational Database Management)

- Open AWS Console → RDS.
- Verify database creation.
- Verify endpoint availability.



STEP 19: LB (Load Balancer)

- Open AWS Console → EC2 → Load Balancers.
- Verify ALB creation.
- Check target group health status.



Troubleshooting:

Issue: Terraform Deployment Failure

Causes:

- Incorrect AWS credentials.
- Syntax errors in Terraform configuration files.
- Resource name conflicts.
- Existing AWS resources with the same name.

Solution:

- Verify AWS credentials using AWS CLI.
- Run terraform validate to check configuration errors.
- Review Terraform error messages and logs.
- Use unique resource names or import existing resources into Terraform state.

Result:

After correcting the configuration and validating the Terraform files, the infrastructure was successfully deployed on AWS.

Conclusion:

This project successfully demonstrated the automation of cloud infrastructure deployment using Terraform on AWS. By implementing Infrastructure as Code (IaC), resources such as VPC, subnets, EC2 instances, load balancer, and RDS database were provisioned automatically rather than being configured manually. The project provided practical knowledge of AWS networking, compute services, and database management. It also highlighted how automation enhances consistency, scalability, and operational efficiency in cloud environments. Overall, this project emphasizes the significance of Terraform in modern DevOps workflows and cloud infrastructure management.